Amirreza Hosseini

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PhD Candidate in engineering management, experienced in complex systems modeling, optimization (Gurobi, CPLEX, Pyomo, MILP, CP, heuristics), reinforcement learning, and decision-support systems. Proficient in key Python libraries, Julia, and R (visualization and process mining). Former financial business analyst with 3 years of experience building cost-revenue and pricing models for payment-industry operations—proven track record of collaborating with teams and translating quantitative models into actionable recommendations for decision-makers.

Areas of Expertise

- Reinforcement Learning
- RL Environments (such as Gymnasium)
- Process Mining
- Feasibility Study
- Operation Research
- Business Analytics
- Programming (Python, Julia, R)
- Version Control Software (Git)
- Optimization Models
- Business Process Management
- Documentation & Reporting
- Statistical Modeling & Visualization

Professional Experience

Stevens Institute of Technology, *Research Assistant* | Hoboken, NJ — *Sep 2023 - Present*

Responsibilities: Conduct Research on Project Scheduling and Extending Hetero-Functional Graph Control Problems. Apply Reinforcement Learning to Plan and Optimize Engineering Systems Operation with a focus on Complex Large-scale Projects.

Project 1. Formulated Petri-net and network flow representations of project scheduling problems as Markov decision processes and developed reinforcement learning algorithms achieving competitive optimization results. (Under Review)

Project 2. Extended the resource-constrained project scheduling problem with hetero-functional graph theory and represented it as a flexible multi-level network flow model with robust validation results. (Under Review)(Conference Paper)

Project 3. Integrated Enterprise Architecture Constraints in Resource-Constrained Project Scheduling with Hetero-functional Graph Theory and solved using Gurobi solvers. (Under Review)(Conference Paper)

Project 4. Introduced a Hetero-functional Graph Theory Perspective of Engineering Management of Mega-Projects (IEEE Access)

Project 5. Collaborated in the development of a Conceptual Introduction to Hetero-functional Graph Theory (Systems Engineering)

Sadad Electronic Payment, *Financial Business Analyst* | Tehran, Iran — *Dec 2020 - Aug 2023*

Responsibilities: Built data-driven feasibility, pricing, and revenue models to support strategic decisions in nationwide payment-industry operations.

Project 1. Built a multi-scenario pricing/feasibility model for a **nationwide gas-station** payment rollout, quantifying FX risk and inflation; delivered tender recommendations adopted by leadership (1% increase in market transactions impact).
Project 2. Modeled the marketing and operational plan of equipping 1 million merchants with variable loan interest rates based on the potential and transactional background of merchants (40% increase in quantity of POS in market)

Education

4.00/4.00 **Ph.D. in Engineering Management**, *Stevens Institute of Technology* | Hoboken, NJ — *2023 - Present*
4.00/4.00 **MSc. in Industrial Engineering**, *Sharif University of Technology* | Tehran, Iran — *2020 - 2023*
3.64/4.00 **B.Sc. in Industrial Engineering**, *Kharazmi University* | Tehran, Iran — *2016 - 2020*

Achievements: Rank 5 @ B.Sc. | Rank 9 @ Nationwide University Entrance Exam

Courses: Reinforcement Learning (Columbia University)| Stochastic Modeling |Discrete Optimization (Coursera)| Decision Analysis | Project Management | BPM | Regression Analysis| Simulation Languages | Introduction to Financial Engineering (Coursera)

Selected Academic Projects:

Project 1. Machine Learning for Stock Market Forecasting – M.Sc. Thesis (Developed classification models using transactional and market data to predict market direction achieving over 70 % accuracy and precision in market direction prediction)
Project 2. Process Mining and Supply Chain Flow Optimization (Applied process mining to activity-log data to reconstruct process flows, identify bottleneck activities, and recommend productivity improvements)
Project 3. Customer Clustering in Payment Industry (Clustered over 2 million users based on application behavior and transaction patterns to support customer segmentation and revenue-oriented strategy design)
Project 4. Feasibility Studies for New Ventures (Conducted feasibility studies for e-commerce and mechanized-agriculture business models, including market, cost, and revenue analysis)
Project 5. Healthcare Patient Classification